ENGINEERING ARCHITECTURE & IMPLEMENTATION SPECIFICATION

Archonita: Title Screen, Multi-Aspect Resolutions & Board Audio HUD

Complete Technical Integration Plan for 'Play', 'Resume Game', 16:9 / 16:10 / 21:9 / 32:9 Aspect Ratios, and Game Board Mute SFX / Mute Music Buttons in Torque2D 4.0

Target Engine: Torque2D 4.0 (Rocket Edition EA4)	**Architecture:** ScriptObject Systems (TORQUE_SCRIPT.md)
Status: Ready for Implementation	**Date:** November 2026

1. Executive Summary & Architecture Principles

This specification establishes the production architecture for modernizing Archonita's user experience across three core pillars: (1) A modular Title Screen UI replacing the legacy full-screen click hijack with dedicated **Play**, **Resume Game**, **Options**, and **Quit** controls; (2) A dynamic Multi-Aspect Ratio and Resolution Engine supporting modern displays (16:9, 16:10, 21:9 ultrawide, and 32:9 super ultrawide) without sprite stretching or hexagon distortion; and (3) An in-game Board Audio HUD featuring independent, persistent **Mute SFX** and **Mute Music** toggles with OpenAL channel isolation.

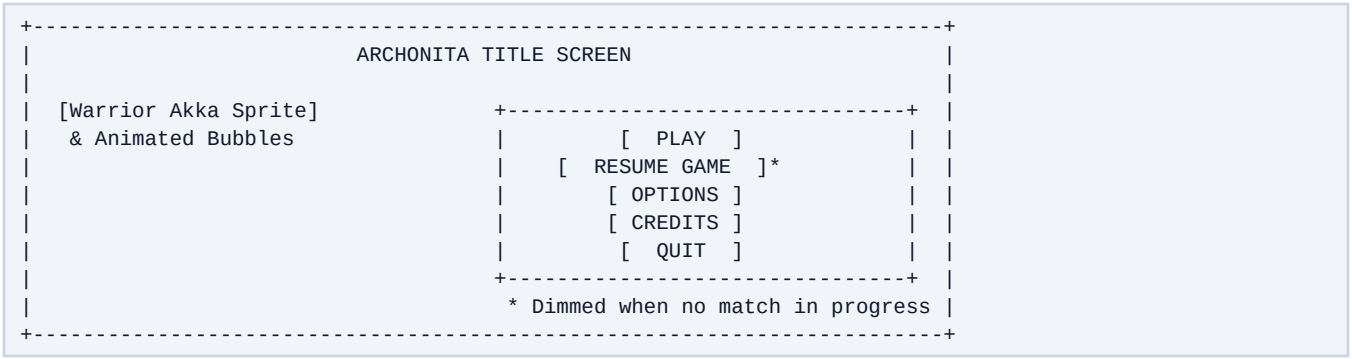
MANDATORY ARCHITECTURE DIRECTIVE (TORQUE_SCRIPT.md):
All new game code MUST follow the strict object-oriented conventions of TorqueScript: one class per file matching the filename, ScriptObject system managers, self-configuration in onAdd, thorough resource teardown in onRemove, and parent-child ownership chains. No global god-namespaces.

2. Title Screen Modernization: Play & Resume Game

Current Defect: In Archon/gameScripts/complexMouse.cs:626-634, any mouse click anywhere inside sceneWindow2D during $gameMainMenuState immediately executes gotoSelectHeroScreen(). There are no UI buttons, no settings access, and no session retention.

Modernized Architecture: We decouple the raw click hijack and construct a dedicated TitleMenuManager (subclassing ScriptObject) that manages an elegant fantasy menu stack:

- **[PLAY] (New Game):** If no session is active, transitions immediately to selectHeroScreen(). If a match is currently in progress, presents a modal confirmation: *'A match is in progress. Starting a new game will discard current progress. Proceed? [Continue] [Cancel]'*.
- **[RESUME GAME]:** Dynamically evaluates $GameSession::hasActiveSession. If *false*, the button is visually dimmed (grayed out) and disabled (setActive(false)). If *true*, it is highlighted and clicking immediately returns the player to board.t2d with intact creature positions, turn counters, hero stats, and deck state.
- **[OPTIONS]:** Pushes the unified Video (Aspect/Resolution) and Audio Settings dialog.
- **[CREDITS] & [QUIT]:** Provides clean navigation to credits and graceful shutdown via shutdownProject(); quit();.



3. Multi-Aspect Ratio & Resolution Engine (16:9, 16:10, 21:9, 32:9)

Mathematical Principle in Torque2D 4.0: SceneWindow camera sizing defines the coordinate system of the 2D world. Archonita originally hardcoded camera dimensions to 100 75 (a 4:3 aspect ratio). When rendered on modern 16:9 widescreen or 21:9/32:9 ultrawide screens, the engine forces the 4:3 camera into the wider window, resulting in severe horizontal stretching (circles become wide ellipses).

Dynamic Aspect Ratio Solution: The vertical camera height is fixed at CamHeight = 75.0 world units to ensure all 9 board rows remain in view. The camera width is computed dynamically from the aspect ratio: CamWidth = CamHeight * (CanvasWidth / CanvasHeight).

Aspect Ratio	Proportion	Camera Size	Supported Resolutions	Target Displays
16:9	1.7778	133.33 x 75.0	1280x720 (HD) **1920x1080 (FHD Default)** 2560x1440 (QHD) 3840x2160 (4K)	Standard PC monitors, Laptops, TVs
16:10	1.6000	120.00 x 75.0	**1280x800 (Steam Deck)** 1680x1050 1920x1200 2560x1600	Steam Deck handheld, MacBooks, Dell UltraSharp
21:9	2.3704 (64:27)	177.78 x 75.0	2560x1080 (UW-FHD) **3440x1440 (UW-QHD)** 5120x2160 (5K2K)	Ultrawide Curved Gaming Monitors (34"-38")
32:9	3.5556	266.67 x 75.0	3840x1080 (Dual FHD) **5120x1440 (Dual QHD)**	Super Ultrawide Monitors (Samsung Odyssey G9 49")
4:3	1.3333	100.00 x 75.0	800x600 1024x768 (Legacy Default)	Legacy CRT monitors, iPad 4:3 fallbacks

Board Framing & Ultrawide Adaptation:

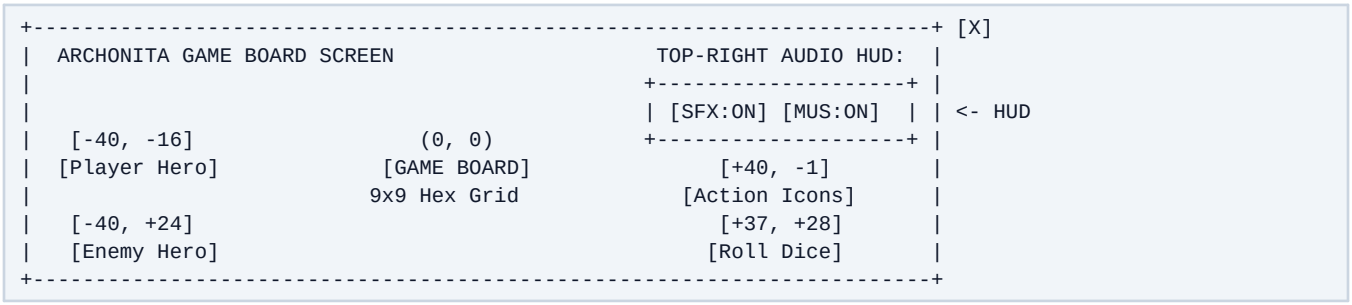
The Archonita 9x9 board sits at world origin (0, 0), spanning 70x70 units. In 16:9, 21:9, and 32:9, the board stays centered without distortion. The background sprite (background in board.scene.taml) is dynamically expanded horizontally (background.setSize(%camWidth + 2, 76)) or flanked by decorative stone border pillars, preventing black voids while maintaining pure geometric fidelity. All GUI overlays use edge-anchored profiles (HorizSizing = 'left'/'right') so HUD elements stay pinned to the monitor edges regardless of aspect ratio.

4. Game Board Screen Audio HUD: Mute SFX & Mute Music

Objective: Provide instantaneous, accessible audio toggles directly within the Game Board match screen without requiring navigation away from gameplay.

UI Presentation & Anchoring:

A dedicated GUI container overlay (managed by BoardAudioHud) is mounted directly to mainScreenGui. By setting HorizSizing = 'left' and VertSizing = 'bottom', the HUD remains firmly pinned to the **top-right corner** of the screen at (Canvas.extent.x - 150) 10 across all aspect ratios and resolutions.



Underlying Audio Subsystem & OpenAL Isolation:

- **Channel 0 (Music):** Controlled via alxSetChannelVolume(0, %vol). When muted, AudioManager::setMusicMuted(true) silences channel 0, pauses/stops %this.bgMusic, and suppresses the looping tick schedule. When unmuted, volume restores to 1.0 and startBackgroundMusic() restarts cleanly.
- **Channel 1 (SFX):** Controlled via alxSetChannelVolume(1, %vol). Additionally, playArchonSound() in t2d_compat.cs and AudioManager::playDiceRollSound() check $AudioManager.isSFXMuted(), avoiding unnecessary OpenAL voice allocation when muted.
- **Visual Feedback:** Unmuted buttons display bright green/gold text ([SFX: ON] / [MUS: ON]). Muted buttons transition to muted slate/red text ([SFX: OFF] / [MUS: OFF]).
- **Hotkeys & Persistence:** Shortcut keys **'M'** (Music) and **'Ctrl+S' / 'N'** (SFX) toggle states instantly. Values persist in $pref::Audio::musicMuted and $pref::Audio::sfxMuted.

5. Game Session Lifecycle & Resume Game Mechanics

Active Match Lifecycle: A global match descriptor $GameSession::hasActiveSession is initiated when BoardManager::showBoardScreen() completes. Pressing Escape during a match summons the modal Pause Menu (**Resume Match**, **Options**, **Restart**, **Quit to Title**).

Pause & Title Navigation: Selecting 'Quit to Title' freezes board schedules and turn timers, flags the session as active, and switches the canvas to the Title Screen. The Title Screen checks $GameSession::hasActiveSession and renders **Resume Game** fully active. Clicking **Resume Game** calls BoardManager::resumeGame(), re-mounting the active t2dScene, unpausing turn timers, and restoring the player's active turn seamlessly.

Session Termination: When a match ends in Victory or Defeat (via CombatEngine.cs), or when the player explicitly confirms starting a New Game, $GameSession::hasActiveSession is cleared to *false*, disabling the Resume Game button.

6. Architectural Inventory: Files to Create & Modify

File Path	Status	Class / Module	Core Responsibility
Archon/gameScripts/TitleMenuManager.cs	**NEW**	TitleMenuManager	Title menu GUI, [Play] / [Resume Game] / [Options] buttons, session queries.
Archon/gameScripts/ResolutionManager.cs	**NEW**	ResolutionManager	Resolution tables (16:9, 16:10, 21:9, 32:9), camera aspect math, 15s safety revert.
Archon/gameScripts/BoardAudioHud.cs	**NEW**	BoardAudioHud	Top-right GUI overlay on board screen, Mute SFX / Mute Music toggles, tooltips.
Archon/gameScripts/PauseMenuManager.cs	**NEW**	PauseMenuManager	In-game Escape pause modal dialog, session freeze/thaw, title navigation.
Archon/gameScripts/OptionsMenuGui.cs	**NEW**	OptionsMenuGui	Video & Audio options dialog (aspect ratio tabs, resolution dropdown, mute sliders).
Archon/main.cs	MODIFIED	Archon Lifecycle	Executes new manager scripts, boots display mode via ResolutionManager, cleanup.
Archon/gameScripts/complexMouse.cs	MODIFIED	Mouse Dispatcher	Decouples full-screen click hijack in $gameMainMenuState; ignores GUI clicks.
Archon/gameScripts/AudioManager.cs	MODIFIED	AudioManager	Adds sfxMuted/musicMuted, OpenAL channel 0/1 volume isolation, preference sync.
Archon/gameScripts/BoardManager.cs	MODIFIED	BoardManager	Mounts BoardAudioHud, dynamic background scaling, match resume handler.
Archon/gameScripts/game.cs	MODIFIED	SceneWindow Loader	Replaces hardcoded 100x75 camera with dynamic aspect calculation.

7. Implementation Roadmap & Execution Phases

- **Phase 1 — Core Audio Engine:** Implement sfxMuted/musicMuted in AudioManager.cs, OpenAL channel volume scaling, and playArchonSound() guards.
- **Phase 2 — Board Audio HUD:** Build BoardAudioHud.cs, configure edge anchoring profiles on mainScreenGui, bind hotkeys 'M' and 'Ctrl+S', verify real-time state sync.
- **Phase 3 — Multi-Aspect Engine:** Build ResolutionManager.cs, implement dynamic camera aspect math, background auto-scaling, and 15s display revert timer.
- **Phase 4 — Pause & Session Management:** Build PauseMenuManager.cs, wire Escape key in gameplay, implement match state freeze and resumeGame().
- **Phase 5 — Modernized Title Screen:** Decouple click hijack in complexMouse.cs, build TitleMenuManager.cs, wire [Play], [Resume Game], [Options], [Quit].
- **Phase 6 — Options Dialog UI:** Build OptionsMenuGui.cs with Aspect Ratio selector, Resolution list, Fullscreen toggle, and Audio controls.
- **Phase 7 — Verification & Conventions:** Execute unit/smoke tests, test resolution matrix across 16:9, 16:10, 21:9, 32:9, audit script conventions via checking-torquescript-conventions.

8. Quality Assurance Matrix & Convention Audit

Rigorous Verification Criteria: (1) *Aspect Ratio Verification:* Confirm camera size matches table exactly and circles remain 1:1 on 1080p, 1440p, Steam Deck 1280x800, and Ultrawide 3440x1440. (2) *Audio Mute Verification:* Validate OpenAL channel volumes and verify dice sound pool / spell audio respect SFX mute without memory leaks. (3) *Resume Game Verification:* Validate board resumption during Player Turn, AI Turn, and Dice Roll without input locking. (4) *Convention Audit:* Every newly authored script will be checked against TORQUE_SCRIPT.md (one class per file, self-configuring in onAdd, clean teardown in onRemove).